

# BTN-1 Macro Deck

Four-key mechanical macro pad for Home Assistant and smart-home control

The Apollo BTN-1 Macro Deck is a four-key control pad with hot-swappable mechanical brown switches, white keycaps, and a status LED in front of each button. Power it over USB-C or an optional battery and connect over Wi-Fi to trigger macros, scenes, media controls, and automations in Home Assistant, Homey, and other smart-home systems. It runs ESPHome and is fully local and open-source.

Each of the four buttons supports single, double, triple, and hold actions — 16 distinct actions total — and an official Home Assistant blueprint makes mapping them quick. Switches are Cherry MX compatible and hot-swappable without soldering, and the board offers extensive GPIO and modular expansion for future add-ons.

## Key Specifications

|                   |                                                                                        |
|-------------------|----------------------------------------------------------------------------------------|
| Product           | BTN-1 Macro Deck                                                                       |
| Manufacturer      | Apollo Automation (Versailles, KY, USA)                                                |
| Inputs            | 4 hot-swappable mechanical switches (Cherry MX compatible)                             |
| Switches included | 4 x brown tactile; white keycaps                                                       |
| Actions           | Single / double / triple / hold per key — 16 total                                     |
| Indicators        | Per-button front status LED                                                            |
| Microcontroller   | ESP32-C6-Mini                                                                          |
| Firmware          | ESPHome — open source (Zigbee in development; Thread/Matter-capable, pending firmware) |
| Connectivity      | Wi-Fi (2.4 GHz)                                                                        |
| Power             | USB-C, or optional 3.7 V 2800 mAh battery (PH 2.0 connector)                           |
| Battery life      | Up to ~1 year at roughly 4 presses/day                                                 |
| Charging          | Onboard battery charges via USB-C                                                      |
| Compatibility     | Home Assistant, Homey, and other platforms                                             |
| In the box        | BTN-1; 4 brown switches; 4 white keycaps; 4 rubber feet; Apollo sticker                |

## Configurations & Pricing

| Configuration                 | Battery              | Price (USD) |
|-------------------------------|----------------------|-------------|
| USB-C (Non-battery)           | —                    | \$34.99     |
| International Battery-powered | Battery not included | \$36.99     |
| USA Battery-powered           | Battery included     | \$39.99     |

International battery model ships with a battery-style case only; source a PH 2.0 cell not exceeding 80 x 30 x 8 mm.

## Features & Use Cases

- **Streaming & macros:** trigger scenes, sound effects, lighting changes, and stream-deck commands for live broadcasts.
- **Physical control panel:** activate Home Assistant scenes, control lights, adjust HVAC, or run routines across platforms.
- **Operations & IoT:** map buttons to machine controls, sensors, or building-automation routines.
- **16 actions:** single, double, triple, and hold on each of the four keys for dense functionality.

- **Hot-swap switches:** Cherry MX compatible — swap switches and keycaps without soldering.
- **Modular future:** GPIO and mezzanine support for e-ink displays, NFC, rotary dials, OLEDs, and sensors.

## Important Notes

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- **Setup via blueprint:** button actions are configured in Home Assistant; the official blueprint walks through single/double/triple/hold mapping.
- **Battery wake delay:** from sleep the device takes ~10 seconds to wake and register a single press; multi-button presses register only while awake.
- **Source your own cell:** for the international battery model, the battery is not included and must fit the PH 2.0 connector and size limit.

## Documentation & Resources

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- **Product page:** [apolloautomation.com/products/btn-1-macro-deck](https://apolloautomation.com/products/btn-1-macro-deck)
- **Getting started:** [wiki.apolloautomation.com/products/btn1/setup/getting-started](https://wiki.apolloautomation.com/products/btn1/setup/getting-started)
- **Home Assistant blueprint:** [wiki.apolloautomation.com/products/btn1/examples/blueprint](https://wiki.apolloautomation.com/products/btn1/examples/blueprint)
- **Source & CAD files:** [github.com/ApolloAutomation](https://github.com/ApolloAutomation)
- **Community:** [Apollo Automation Discord](#)